Progress Report: Muscle vs. Movement Mismatch in Children's Walking

Project for Confluence'27 (IEEE) | Status as of 31 October 2026 (update 2)

1. What the project is about

Children with **cerebral palsy (CP)** or **idiopathic toe walking (ITW)** walk differently from other children. We compare two things for each child: **how the ankle moves** (what we can see) and **when the shin and calf muscles switch on** (measured with EMG sensors on the skin). We want to find cases where these two disagree, for example the ankle looks fine while the muscles are still firing at the wrong time. Because a child never walks exactly the same way twice, every result is checked against the normal difference between two walks recorded on the same day.

2. Data used

Public dataset **LIVE-GaitNeuroKids** (Zenodo, v1.1). Only the already-processed files were used: ankle angle, shin (tibialis anterior) and calf (gastrocnemius) EMG, and foot-on-ground labels, already split into individual steps.

Group	Children	Walking files used
CP (cerebral palsy)	19	91
ITW (toe walking)	11	39
TD (typically developing, the 'normal' reference)	25	49

3. What we did, step by step

- **Loaded all files.** 179 files loaded. *Problem:* 19 files contain the ankle angle for only one leg. *Fix:* the missing leg is skipped and logged; the other leg is kept.
- **Checked the before/after visits.** *Problem:* many children were treated more than once and the visit labels (T0 = before, T2 = 3 months after) restart for each treatment, so most 'before/after' pairs mixed two different treatments. Only 7 children have a valid before/after pair. *Fix (approved):* the main analysis looks at every visit on its own; the before/after comparison becomes a small pilot.
- **Built the 'normal walking' curves** from the 25 TD children (each child counted once). They match textbook walking: calf active while the foot is on the ground, shin active when the foot is in the air (Figure 1).
- **Scored every visit**: how far the ankle curve and each muscle curve are from normal, separately for the foot-on-ground (stance) and foot-in-air (swing) parts of the step.
- **Measured the normal wobble** between two walks on the same day, to know which differences are real.
- **Sorted each visit into four boxes**: both normal, ankle-only abnormal, muscle-only abnormal, or both.
- **Robustness check:** repeated everything after removing a possible ankle-sensor offset (a 'crooked ruler' effect), so only the shape of the ankle movement counts.

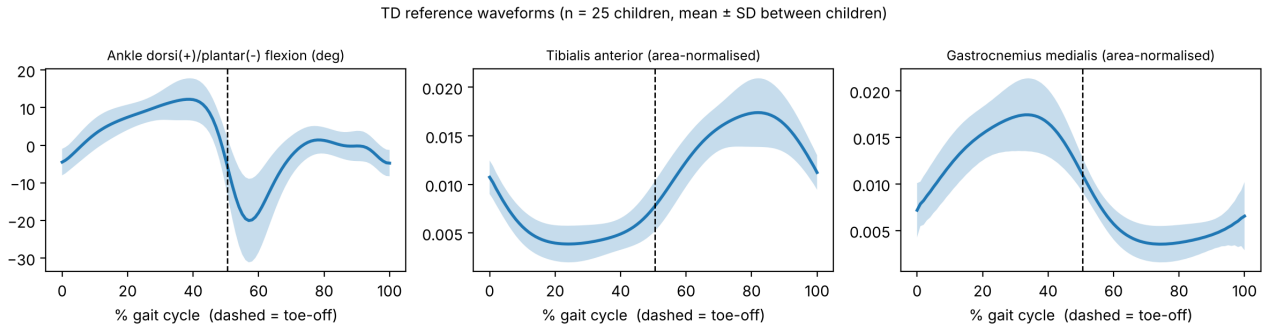


Figure 1. Normal reference curves from 25 TD children (line = average, band = spread between children, dashed line = foot leaves the ground).

4. Main results

Group	Ankle abnormal, muscles normal	Muscles abnormal, ankle normal	Shape-only check (ankle offset removed): ankle-only / muscle-only
TD (background)	10%	8%	6% / 8%
CP	**53%** (95% CI 31-71)	8% (2-17)	**44%** / 13%
ITW	2%	**19%** (9-30)	0% / **21%**

- **Even healthy children land in an 'abnormal' box about 20% of the time** just by chance, so every patient number must be compared with that background, not with zero.
- **CP: the ankle movement is the problem, not the muscle timing.** About half of CP visits show abnormal ankle movement with normal-range muscle timing (vs 6-10% in healthy children). This holds after removing sensor offset (44%), but the strictest version drops from 42% to 13%, so part of it was sensor placement.
- **ITW: the opposite pattern.** About 1 in 5 visits show abnormal muscle timing with a normal-looking ankle, mainly the calf staying active while the foot is in the air, which fits toe walking. This is a trend: the confidence interval overlaps slightly with healthy children.
- **Muscle-timing differences are smaller than same-day wobble.** With the strictest definition no muscle-only cases remain in any group.
- **Before/after pilot (7 children):** after removing sensor offset, only 1 child (a CP child) shows 'ankle improved, muscles did not'. No child's muscle score ever improved beyond the wobble, so this pilot cannot show treatment effects on muscle timing.

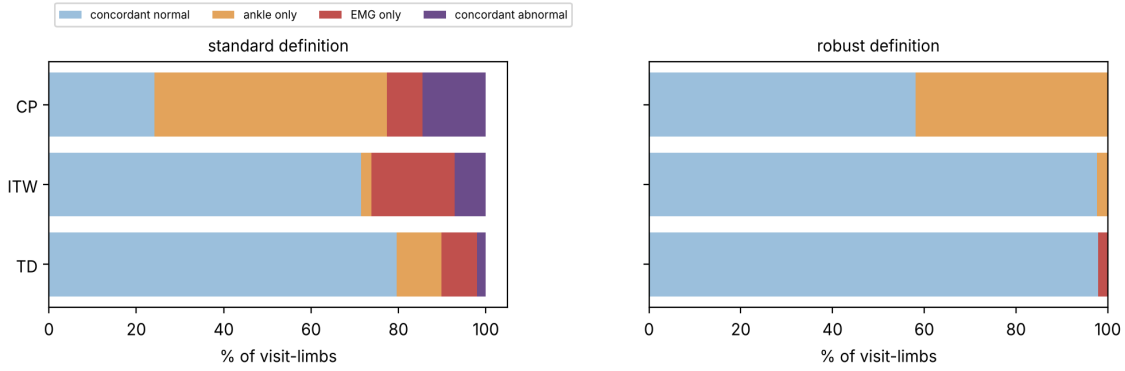


Figure 2. Share of visits in each box. Left: standard definition. Right: strict definition (must also exceed same-day wobble).

5. Extra checks before writing (completed)

Before writing, we tested whether the results could be explained by luck, by a few children with many visits, or by age (the healthy children were younger on average).

Check	What we asked	Result
Direct comparison with healthy children	Is the gap real, or could it be luck? (shuffle test, 5,000 times)	CP ankle-only: **+43 points, p = 0.001** (real). CP muscle-only: no difference. ITW muscle-only: +11 points, **p = 0.18** (not significant).
One vote per child	Do children with many visits drive the numbers?	No. CP ankle-only 48% per child vs 53% per visit; TD 10% either way.
Age	Could age explain the differences?	No. In healthy children age is unrelated to every score. With age-matched groups (about 8-9 years each) the CP result gets stronger: 60% vs 10%, p = 0.0005.
Sensor offset (repeated)	Do the above still hold with shape-only ankle scores?	Yes for CP (+37 points, p = 0.003). ITW still not significant (p = 0.13).

6. What we can now claim

Claim	Status
CP: abnormal ankle movement while muscle timing stays in the normal range	**Solid**: passes every check
CP: muscles abnormal while ankle looks normal	**Not supported**: same rate as healthy children

ITW: calf active while the foot is in the air	**Trend only**: needs a larger study
Muscle-timing differences are smaller than same-day wobble	**Solid**: holds everywhere
Before/after pilot	**Descriptive only**: 1 of 7 children after the offset check

Main message of the paper: in CP, ankle-movement problems are common while the shape of muscle activation stays within normal and trial-to-trial variability. Muscle-timing differences are mostly smaller than the normal difference between two walks on the same day, which currently limits the use of EMG shape to track treatment response. This is a change from the original idea (finding children whose ankle improves but muscles do not), because the data showed that pattern cannot be measured reliably here.

7. Remaining limitations and next steps

- The muscle measure averages over a whole phase, so short timing errors may be missed (to be stated in the paper).
- Small numbers: 11 ITW children and only 7 valid before/after pairs.
- One visit is labelled differently in the file name and the participant sheet; the detected 'foot leaves the ground' point varies widely (35-81% of the step).
- **Next:** agree on the main message, then write the 6-page IEEE paper. All code is in a git repository and every result can be re-run from the raw files.